

Healthcare Benefits Management Platform

A HIPAA-compliant, multi-tenant SaaS platform powering benefits administration, advisor management, and member engagement — built on a modern TypeScript monorepo.

4

Applications

15+

Shared Packages

14

Edge Functions

40+

Database Tables

12+

Integrations

SYSTEM PURPOSE

What MPB Health Does

An end-to-end digital platform replacing legacy tools like Zoho CRM with a unified, healthcare-specific ecosystem for benefits administration.



Benefits Administration

Member enrollment, claims management, document handling, and payment processing — all in one HIPAA-compliant platform.



CRM & Sales Pipeline

Full-featured CRM with lead scoring, deal pipelines, quoting, invoicing, and AI-powered insights — replacing Zoho end-to-end.



Advisor Command Center

Training LMS, video meetings, playbooks, lead management, and engagement tools for a distributed advisor network.



Multi-Tenant Organizations

Champion system enabling white-label organization management with billing, compliance AI, analytics, and engagement inbox.



Compliance & Audit

HIPAA-grade security with PHI access logging, tamper-proof audit trails, MFA enforcement, and encryption key management.



Content & Marketing

Blog CMS with AI generation, resource library, bulletin system, form management, and multi-channel analytics tracking.

TECH STACK

Built with Modern Tools

A carefully chosen stack optimized for developer velocity, type safety, and healthcare compliance.

◆ FRONTEND

- React 18
- TypeScript 5.5
- Vite 5
- React Router 7
- Tailwind CSS 3
- React Hook Form
- Zod 4
- Framer Motion
- Zustand
- Recharts
- TipTap Editor

◆ BACKEND & DATABASE

- Supabase
- PostgreSQL 17
- Supabase Auth
- Supabase Realtime
- Supabase Storage
- Edge Functions (Deno)
- Row-Level Security

◆ DEVOPS & TOOLING

- Turborepo
- pnpm 9
- Vercel
- Vitest
- ESLint
- Changesets

◆ AI & AUTOMATION

- Google Gemini API
- N8N Workflows
- AI CRM Agent
- AI Blog Generation
- Compliance AI

APPLICATIONS

Four Apps, One Monorepo

Each application serves a distinct user persona while sharing packages for consistency and velocity.

● Marketing Website

`www.mpb.health`

- Member enrollment & onboarding
- Claims management portal
- Document & payment management
- Blog CMS with AI generation
- Multi-channel analytics (GA4, Pixel, Clarity)

● CRM Platform

`crm.mpb.health`

- Lead scoring & deal pipelines
- Account & contact management
- Quoting, invoicing & payments
- AI-powered CRM agent
- Zoho CRM bi-directional sync

● Advisor Portal

`advisors.mpb.health`

- Training LMS & progress tracking
- Jitsi video meetings
- SOP playbook library
- Lead management & content hub
- Bulletin & notification system

● Admin Portal

`admin.mpb.health`

- User & enrollment management
- Content management (blog, resources)
- Payment processor configuration
- Security audit logs
- System settings & SMS accounts

mpbhealth-monorepo/

apps/

website/	# www.mpb.health — React + Vite
crm/	# crm.mpb.health — React + Vite
advisor-portal/	# advisors.mpb.health — React + Vite
admin-portal/	# admin.mpb.health — React + Vite

packages/

@mpbhealth/ui	# Shared React component library
@mpbhealth/database	# Supabase client & TypeScript types
@mpbhealth/auth	# Auth services, MFA, security events
@mpbhealth/config	# Feature flags, constants, env config
@mpbhealth/utils	# Validators, sanitizers, helpers
@mpbhealth/crm-core	# CRM business logic
@mpbhealth/champion-core	# Multi-tenant champion system
@mpbhealth/integrations	# External API adapters
... 7 more shared packages	

supabase/

migrations/	# PostgreSQL schema migrations
functions/	# 14 Deno Edge Functions
turbo.json	# Turborepo pipeline config
pnpm-workspace.yaml	# pnpm workspace definition

Data Flow Diagram

Request lifecycle from client to database and back, with security and compliance at every layer.

CLIENT LAYER

Website

www.mpb.health

CRM

crm.mpb.health

Advisor Portal

advisors.mpb.health

Admin Portal

admin.mpb.health



EDGE / CDN LAYER

Vercel Edge Network

Global CDN + Security Headers



AUTHENTICATION LAYER

Supabase Auth

JWT + MFA + Rate Limiting



RBAC Engine

5 Roles + RLS Policies



API & BUSINESS LOGIC LAYER

Supabase REST API

Auto-generated from schema

Supabase Realtime

WebSocket subscriptions

Edge Functions

14 Deno serverless functions



DATA LAYER

PostgreSQL 17

40+ tables, RLS on all

Supabase Storage

Documents & media



EXTERNAL SERVICES

Stripe / Square

Payments

Resend / Mailchimp

Email

Twilio

SMS

Gemini AI

Intelligence

Zoho CRM

Sync

HIPAA-Grade Security

Defense-in-depth architecture with compliance baked into every layer — not bolted on after the fact.



Network & Transport Security

All traffic encrypted via TLS 1.3. Vercel Edge enforces HSTS, CSP, X-Frame-Options, and CORP headers on every response. Strict Referrer-Policy and Permissions-Policy applied globally.

HSTS

CSP

TLS 1.3

X-Frame-Options

CORP / COEP



Authentication & Identity

Supabase Auth with multi-factor authentication (TOTP & SMS). Brute-force protection via rate limiting, login attempt tracking, and device fingerprinting. Password history enforcement prevents reuse.

MFA (TOTP + SMS)

Rate Limiting

Device Fingerprinting

Password History

Session Management



Authorization & Data Isolation

Role-Based Access Control with 5 tiers (super_admin through guest). PostgreSQL Row-Level Security on every table. Multi-tenant data isolation via org_id scoping ensures zero cross-tenant leakage.

RBAC (5 Roles)

Row-Level Security

Multi-Tenant Isolation

org_id Scoping



HIPAA Compliance

PHI access logging with dedicated audit table. Encryption key management for sensitive fields. HIPAA mode toggle per organization. Message approval workflows for compliance review. BAA-covered Supabase infrastructure.

PHI Access Log

Encryption Keys

HIPAA Mode

Approval Workflows

BAA Coverage



Audit & Monitoring

Tamper-proof security event log with cryptographic hash chaining. Real-time security alerting via webhooks (Slack, PagerDuty). Comprehensive login attempt and session tracking across all portals.

[Hash-Chained Audit Log](#)[Security Webhooks](#)[Login Tracking](#)[Alert System](#)

ECOSYSTEM

Integration Network

Pluggable architecture connecting to best-in-class services across payments, communications, analytics, and AI.

Stripe

Payments

Square

Payments

PayPal

Payments

Authorize.net

Payments

Braintree

Payments

Resend

Transactional Email

Mailchimp

Marketing Email

Twilio

SMS

Vonage

SMS

MessageBird

SMS

Zoho CRM

CRM Sync

Jitsi Meet

Video

Google Gemini

AI

N8N

Automation

Cognito Forms

Forms

Google Analytics

Analytics

Facebook Pixel

Tracking

Microsoft Clarity

Heatmaps

Hotjar

UX Analytics

LinkedIn Insight

B2B Tracking

TikTok Pixel

Social Tracking

Pinterest Tag

Social Tracking

Twitter Pixel

Social Tracking

Zoho SalesIQ

Live Chat

INFRASTRUCTURE

Built to Scale

Architecture decisions that ensure the platform grows with the business — from hundreds to hundreds of thousands of users.



Edge-First Delivery

Vercel's global Edge Network serves static assets from 100+ PoPs worldwide. Sub-50ms TTFB for all client applications with automatic ISR and CDN caching.



Serverless Compute

Supabase Edge Functions scale to zero and auto-scale on demand. No server provisioning, no cold-start management — pay only for execution time.



Managed PostgreSQL

Supabase-managed PostgreSQL 17 with automatic backups, point-in-time recovery, connection pooling via PgBouncer, and read replicas available on demand.



Real-Time at Scale

Supabase Realtime handles millions of concurrent WebSocket connections for live data updates across CRM, advisor, and admin portals.



Monorepo Efficiency

Turborepo with remote caching ensures CI builds only rebuild what changed. Shared packages eliminate code duplication across 4 applications.



Multi-Tenant Architecture

Champion system's org_id-based isolation scales horizontally. Each tenant's data is logically separated at the database level via RLS policies.

Architecture Principles

Type Safety End-to-End

TypeScript from UI to database types. Supabase CLI generates types directly from the PostgreSQL schema, ensuring compile-time safety across all 4 apps and 15+ packages.

Shared Package Architecture

Business logic lives in domain packages (@mpbhealth/crm-core, champion-core, etc.), not in app code. Apps are thin presentation layers consuming shared services.

API-First Design

Supabase auto-generates REST and GraphQL APIs from the database schema. Edge Functions handle complex integrations. Zero custom backend servers to maintain.

Progressive Enhancement

Configurable payment processors, SMS providers, and CRM integrations. Feature flags enable gradual rollouts. The platform adapts to business needs without re-architecture.

MPB Health

Technical Architecture Overview — Confidential

Built with React · TypeScript · Supabase · Vercel · Turborepo